High-pass filters

Launcher: `Toolbox/highpass.bat`

Purpose

High-pass filters suppress gradual background variation and emphasize local edges or texture. They are useful for exploring faint linear, rectilinear, circular, and textural patterns, but also enhance roads, plough lines, field boundaries, image seams, and noise.

Inputs and parameters

Option	What to provide	Behaviour
Input image	Any GDAL-readable raster, normally GeoTIFF	Only one selected band is processed.
Image band to process	Positive one-based band number	Default: `1`. It must exist in the source raster. For a derived single-band index or PCA component exported separately, use `1`.
Results folder	Existing folder, or blank	Blank creates `HIGHPASS` in the input image's grandparent directory. A manually selected folder must already exist.
Vector layer	Optional clip polygon	Crops all five results after filtering.
Output compression	`Lossless` or `Near-lossless`	Lossless (default) is bit-exact. Near-lossless (LERC) is about 2.5× smaller, with an error of at most 0.1 % of each filter output's value range (99th minus 1st percentile). See Output compression.

Filters and outputs

For input `scene.tif`, the tool creates:

Output suffix	Operation	Reading the result
`_laplacian.tif`	3×3, four-neighbour Laplacian	Highlights rapid change in all directions; positive and negative values mark opposite sides of edges.
`_log5.tif`	5×5 Laplacian-of-Gaussian-style kernel	Uses a broader neighbourhood and is somewhat less sensitive to single-pixel variation.
`_sobel_mag.tif`	Magnitude of horizontal and vertical 3×3 Sobel derivatives	Non-negative edge strength independent of direction.
`_highboost.tif`	3×3 high-boost sharpening	Retains the original signal while strongly sharpening local detail; useful for visual inspection, not direct reflectance measurement.
`_glcm_contrast.tif`	Local 7×7 contrast using 32 grey levels and four directions	High values indicate locally dissimilar neighbouring pixels and textural boundaries.

All results are single-band Float32 Cloud Optimized GeoTIFFs, retain georeferencing, use DEFLATE (lossless) or LERC (near-lossless) compression, and preserve source NoData where available. If the source NoData is `0`, near-lossless outputs can turn valid values very close to 0 into NoData; use lossless in that case. Computation is tiled for large rasters. Existing files with the same names are overwritten by the filtering operations.

Interpretation

Display Laplacian and LoG outputs with a diverging colour ramp centred near zero. Use a sequential ramp for Sobel magnitude and GLCM contrast. Compare features at multiple display scales and always check the unfiltered source. A strong edge response shows contrast, not the cause or age of that contrast.